

Exercise 10: Web Dashboard Layout

Three notebook sections, three captions and a reflection. No separate document.

Item	Details
Points	100
Starting point	Your Exercise 9 Observable notebook, duplicated as DIV 6380 Exercise 10
Data	clean/mvu_student_dim.csv (already in your notebook)
Resources	MDN: CSS grid layout and flexbox. Observable: Inputs. WebAIM contrast checker. Learning Tableau 2025, Ch 8 (dashboards) for the Tableau side, free through UTD Libraries.
Walkthrough	Module 10 of the MVU Tableau Walkthrough has every code cell, ready to copy, with checkpoints.
Submit	A link to your Observable notebook that your instructor can open.

Before you start

In Exercise 5 you laid out a Tableau dashboard with containers. Here you do the same job with CSS: flexbox for a row of items (a horizontal container), a 12-column grid for rows and columns together, CSS variables for the theme, and a media query for small screens (Tableau's device layouts).

Same two editors as Exercise 9. The walkthrough code uses Notebooks 2.0 style. In the older editor, drop `const` and use `viewof` for inputs. The `html`...`` template works in both.

Section 1: headline numbers in a flex row

1. Duplicate your Exercise 9 notebook. Keep the data, cohorts, minGPA and filtered cells.
2. Compute kpis: students (filtered.length), graduation rate, withdrawal rate and average aid, all from filtered.
3. Build kpiRow: four cards in a flex container (display: flex; flex-wrap: wrap; each card flex: 1 1 180px).
4. Make the cards say which cohorts they cover, for example with a subtitle that reads the cohorts input.

Checkpoint: With cohorts 2020 and 2021 and minimum GPA 2, what does the Graduation rate card read?

Section 2: two charts in a 12-column grid

1. Turn your Exercise 9 chart into a reusable function, `barChart(rows, options)`, with a `viewBox` and no fixed width. Add a helper, `rateBy(rows, key, field)`.
2. Write one style cell: a 12-column grid (repeat(12, 1fr), gap 20px), a KPI row spanning 12, a main chart spanning 8 and a side chart spanning 4.
3. Build the dashboard cell: title, KPI row, graduation rate by college (main), withdrawal rate by enrollment intensity (side).
4. Try spans of 6 and 6, then choose the split you'll defend.

Checkpoint: Part-time withdrawal rate in the side chart, 2020–21 cohorts?

Section 3: theme, contrast and phones

1. Move colors into CSS variables on .dashboard: --brand (UTD green #154734), --accent (UTD orange #e87500), --ink, --muted, --card. Replace hard-coded colors with var(--...).
2. Check UTD orange against white and against the card gray in the WebAIM contrast checker. Use the accent only where it passes for that use.

Checkpoint: Contrast ratio of UTD orange on white, to one decimal?

3. Add a media query that stacks the charts below 700px wide. Test by narrowing the window or on your phone.
4. Decide what to do about chart labels that become too small on a phone, and do it.

Captions and reflection

Cell	What it must answer
Section 1 caption	How do your KPI cards tell the reader which cohorts they cover? What would a reader get wrong without that?
Section 2 caption	Why did you give each chart the width you did? What did turning the chart into a function make easier?
Section 3 caption	Where did you use the accent color, and where did the contrast check stop you? What did you do about the phone layout, and why?
Reflection	Compare CSS grid and flexbox with Tableau's layout containers. What maps across directly, what's easier in each, and which would you choose for an executive dashboard that must work on phones? About 250 words.

The reflection carries 25% on its own. "CSS is more flexible" does not earn full marks. Name specific equivalents (flex for a horizontal container, grid spans for distribute-evenly, media queries for device layouts) and say where each tool helped or got in the way.

Before you submit: open your notebook link in a private browser window, and check the layout at phone width.

Submit for Exercise 10

- A link to your Observable notebook, openable by your instructor
- Section 1: four KPI cards in a flex row, reading the inputs, labelled with the cohorts they cover
- Section 2: a 12-column grid with the KPI row and two charts from one reusable chart function
- Section 3: CSS variables for the theme, a contrast-checked use of the accent, and a media query that stacks the charts
- A Markdown caption after each section, and the reflection at the end